

Product design

1. Why might an actuary be looking at a possible new design?
- External and Regulatory Drivers –
 - Changes in legislation - increase the capital requirement making them too expensive to run.
 - Change in Tax - tax advantage.
 - competitors - protect its market share.
- Awareness and inadequacy in current prod –
 - Insufficient profitability after a while
 - **New business Capital strain on day 1. Alternative are required**
 - Risk exposure
- Shifting consumer needs - Inflation protection
- Demographic changes - Aging population needs retirement prod
- Operational efficiency - exploiting internal expertise
- distribution efficiency – shift from expensive ind brokers to digital portals or banking

FACTORS IN PRODUCT DESIGN – DUCC FORCEDD RAMPSSS

O – Onerousness of any guarantees

- Burden of meeting any guarantees Problems with guarantees – suffer a cost that was not expected, level of reserve need to keep
- Burden will depend on – guarantee size, period, significance of reserve, volume of business, available capital
- Types of guarantees – SV, mortality, interest, bonuses, expense, annuity option, renewability w/o medical
- **Burden of guarantees can be reduced by reducing the financial impact like by backing the liability by matching investment, for eg SV – secure short term investment like govt bonds.**

R – Risk characteristics and risk manage

Acceptable level of risk associated with the proposed product design. Depend on the ability and willingness to either absorb or reinsure to hedge it.

Companies will consider, what can go wrong, financial impact in the worst case. Company can do below to manage –

- **Offer UL or reviewable contracts to avoid long term rate guarantees**
- **Reinsure**
- **Incorporate ample margins in the premium rates**
- **Offer the contract as an additional ‘rider’**

M – Marketability

- **Benefits offered must be attractive to the market.**
- **Innovative design, addition to options, tax advantage and guarantees**
- **Inline with the need and/or demand of market which can be assessed by surveys**

- Premiums should be in regards to benefits

S-Sustainable investment options

- Increasing focus is being placed on ESG (Environmental, social and governance)
- The drivers for these are the combination of regulatory, marketing and social responsibility (working condition of employees)
- A company needs to consider the sustainability of investment options when designing products.

Interaction of product design factors

Factors stated above are not necessarily independent also they are not necessarily exclusive

2. List the pair of factors that are likely to be difficult to reconcile

⇒ To maximise the below mentioned factors later factors have to be compromised

Financial/Capital requirement, Onerous guarantees, Profitability, Sensitivity of profit, risk characteristics
Vs

Marketability, competitiveness, cross-subsidise

Pair 1: Profitability vs. Competitiveness

- For **Profitability**:, thick safety margins into the premium - expensive premium or lower guaranteed benefits, which ruins the product's **competitiveness**

Pair 2: Onerous Guarantees vs. Marketability

- Offering a high guarantees makes a product highly **Marketable** and easy to sell. But becomes heavily **Onerous Guarantees** that require massive statutory capital backing and create severe insolvency risks if the economy declines.

Pair 3: Financing Requirement vs. Marketability

- **The Tension:** Distribution channel more commission -> **Marketable** increases the company's **Financing Requirement** (New Business Strain).

Pair 4: Sensitivity of Profit vs. Competitiveness

- **The Tension:** To make a unit-linked or index-linked product highly **Competitive**, the company might choose to charge fees based strictly on a percentage of the assets under management (FUM). However, this means if the stock market crashes, the company's fee income collapses instantly. By chasing competitiveness, you have drastically increased the **sensitivity of corporate profit** to market shocks.

Pair 5: Cross-Subsidies vs. Competitiveness

- **The Tension:** As we discussed with simple charging structures, a company might use a flat percentage fee to keep the product simple and marketable. This creates a **Cross-Subsidy** where large policies pay for small policies. The problem is that a precise competitor will identify your overcharged large-policy clients, offer them a cheaper rate, and ruin your **competitiveness** in that highly profitable segment.

Product design examples

Unit linked pensions – pay value on retirement or earlier death or Surrender

Setting Assumptions

Assumption setting framework

- Consider the other required factors based on purpose:
 - Solvency Capital requirement
 - Profit margins, USP (Unique selling proposition), Loss leader, Competition, Risk appetite, data quality, Free assets
- Compare the change in assumption from last time and explain (e.g. – in reserving and re-pricing)

Mortality trends –

- Assumption needs to reflect the expected future mortality of relevant lives and therefore consideration should be paid to the expected changes in rates over time.
- Mainly impact the annuity where longevity increases similarly for contracts with significant sum assured, any expectation of increasing mortality should be included in the basis.
- For some contracts, like single premium contracts, it can simply be ignored.
- More important, if premiums are non-reviewable.
- Approaches –
 - Expectation approaches – involve expert opinion and subjective judgement on how mortality will improve over time.

ADV - All relevant knowledge is included.

DIS – expectations are subjective and could be biased.

- Extrapolation approaches – projecting historical trends and experience in future. some elements of subjective judgement, eg, choice of period over which such trends are to be determined

DIS – Complex. Difficult to get a reliable assumption especially when historic experience is volatile to a very particular extreme event.

- *Explanatory approaches – model the mortality trends from a biomedical perspective. effective to the extent that the processes causing death are understood and can be mathematically modelled.*

Four major causes of death are – cancer, circulatory disease, respiratory disease, and infectious .

Understanding how the mortality due to these causes have changed and will continue to change helps with mortality projections.

Multi-factor predictive modelling techniques are increasingly in use. These techniques can combine internally held customer data with external drivers (medical advancement), allowing for correlations and interactions between them.

Two methods of stochastic modelling – Lee-Carter and P-spline

Investment return

- Purpose – pricing and best estimate
- Determine the significance of the assumption, i.e., impact on the profitability of the contract, which will depend on the level of reserve built up and investment guarantees given.
 - More sensitive- > more margins of uncertainty -> More reserve -> more investment income.
 - Higher investment guarantees, greater care over setting the assumption.
- This will also impact the type of assets in which the premiums from the contract will be invested.

- The extent of *reinvestment risk* of reinvestment of future cashflows like - future premiums of IN or NP or proceeds from investment and to which it can be reduced by a suitable choice of assets. The best estimate return available from future investment may well differ from the same available for investment now and may also differ how far into the future the investment is made. Therefore, the overall best estimate of investment assumptions will reflect the expected balance between the expected future and current investments' yields..
- Consider the likely mix of assets which will back the contract in the future, investigate the yield now, and attempt to predict the future returns bearing in mind the impact of future changes to the economic environment. Intended investment mix for the contract will not be derived from the contract looking in isolation, as it will be affected by the level of free assets available to the company to support writing the business. This is because more free assets, less need to match the assets exactly.

Market Consistency

If a market consistent approach is used then expected investment return can be set as the risk-free.

However, if the stochastic approach is adopted for market consistency, then the investment return volatility and correlation assumption remain dependent on the actual underlying asset types.

Allowing for tax and future bonuses

If a fund is subjected to tax on returns, then adjustment for that also needs to be made, involving applying appropriate rate of tax to the different components of Inv Return and income.

Some sort of assumption on future bonuses must be made by company and included in the premium. For this conservative assumption or realistic assumption with explicit allowance for bonus is used.

Expense and Commision

- Purpose – Pricing and best estimate
- *Marginal expenses - initial acquisition, initial medical underwriting, initial administration, renewal administration, renewal reward to sales channel, investment, withdrawal, claim*
- SAME AS MORTALITY
- Analyze the data – result of analysis will be divided in expense by function, premium & benefit
- Adjustments –

To reflect the expected expenses to be incurred in processing and subsequently administering the business to be written under the product being priced.

- Include per policy expense – might have thought of the expense loading, but will vary in practice in proportion to the size of the policy being taken out. However it is a problem because, large policies are subsidizing small policies which goes against the principle of equity and possibly making them uncompetitive. This can be coped by:

- *Individual calculation of premium rates - This could only be done for large cases because small policies could end up paying a very large percentage of their premium as expense loading.*

- *Policy fee addition to the premium – or deduction from regular benefit payments for non-linked or charges that match per policy expense for unit linked. With modifications for competitiveness as inc in percentage of expense contribution. However this modification won't work if more small policies are sold more than anticipated.*
- *'Sum assured differential'. For NL, different premium rates are charged according to which band the benefit requested falls into. For UL, different charges are applied according to which band the premium payable falls into.*
- *Commission rate will depend on the target market*
 - Following can be considered when setting the expense inflation parameters –
 - Current rates of inflation, both for prices and earnings
 - Expected future rates of inflation
 - Must also consider two different aspects of expense inflation:
 - Expense inflation during the term of future new policy and
 - The expense inflation between 'now' and the dates at which the future new policies are actually issued.

Margins

- By including margins somewhere in the assumption, the risk from adverse future experience is reduced.
- Approach 1 – Margins applied on the expected values.
For example – $i' = i'' - m$ where i'' is the best estimate of the future annual investment, m is the margin and i' is the interest rate assumption in the pricing basis => larger the value of m , lower that assumed interest=> lower the marketing loss
- Approach 2 – The risk element of risk discount rate - individual best estimate + higher RDR
- Approach 3 – using a stochastic approach. All parameters are best estimates and Risk discount rate = risk free discount rate. Then we plug probability distribution and run thousands of stimulations. These stimulations give distribution of premiums based on different market conditions. Then we pick the safe one from that range. The risk to the company will be that the actual experience is in line with one of the more extreme outcomes of the distribution.
- In pricing, margins will also be influenced by the need to have competitive premiums.
- Size of the margins must crucially depend on:
 - the degree of risk associated with each parameter
 - used the financial significance of the risk from each parameter.

Discount Rate

Risk Discount Rate

- Net future expected cashflows are discounted back @RDR to get NPV.
- Investors or providers of capital expect a return at a certain discounted rate which is generally more than the safe investment as compensation for the risk of default, commercial failure and so on. This part is known as 'risk premium'. And risk free DR + risk premium = risk discount rate.
- What makes a product risky –
Lack of historical data, High guarantees, Policyholder options, Overhead costs, Complexity of design, Untested market

- Product features that might lead to a lower risk discount rates –
 - Benefits based on contingencies about which there is a lot of reliable data
 - No guarantees
 - No PH options
 - Low overhead costs (since this reduces the risk posed by low business levels)
 - Simplicity of design (and so of understanding, communicating, pricing, and administering)
 - Using a tried and tested market
 - The ability to vary charge and/or review premiums.
- *The level of statistical risk could be assessed by using –*
 - *by considering variance of the individual parameter values used*
 - *Sensitivity analyses with deterministically assessed variation in the para values as it tells us what changes in our assumption are more profitable.*
 - *By using stochastic models for the parameter values and stimulation.*
 - *By comparison with any available market data*

small variance low risk discount rate and more variation => can be quite sensitive to economic swings. So, we would use a higher risk discount rate for valuing.

CAPM

- The idea behind the CAPM is that a well-diversified portfolio of shares cancels out specific risk and leaves only systematic.
- It is possible to derive the following formula which expresses the expected return on any asset in terms of the market return. This is given by: $E_i = r + (E_m - r) \beta_i$ where: E_i is the expected return on asset i (eg our company's shares) r is the return on a risk-free asset E_m is the expected market return (eg the stock market) β_i is the 'beta factor' of asset i (defined as the covariance of r_i and r_m divided by the variance of r_m).
- $E_i - r$ is the risk premium of an asset and $(E_m - r)$ is the risk premium of the market
- β_i is the measure of riskiness of the asset relative to that of the market.

1. *Describe how a risk discount rate could be determined that reflects the expected level of statistical risk, for a given new product, including the use of the following:*

- a) *Deterministic sensitivity analysis*
- b) *Stochastic simulation*

It can be assumed that the SH's required return on capital is known.

- *Calculate the trial premiums of the product based on the best estimated assumption of future experience, using the SH's required return as the risk discount rate.*
- *Calculate the return on capital generated by these premiums, on the basis of a range of different deterministic scenarios for the future experience. The scenarios used should include adverse outcomes that have a feasible likelihood of occurring.*
- *Should the return on capital obtained fall below some required minimum level, then the premium should be increased until this criteria is satisfied and if return is in excess, then premium can be reduced.*
- *Repeat step 2 and 3 to model the variables stochastically. Calculate the frequency by which the return on capital falls below the required minimum rate. If the frequency is high, then the premiums could be further raised until a 5% frequency is obtained.*
- *Calculate the return on capital obtained from any new level of premium produced from steps (2, 3, 4) using best estimate experience assumptions.*

- *This new return on capital can then be used as the new risk discount rate which takes account of the levels of statistical risk for the product.*
- *In case of repricing, start with the current risk discount rate at stage 1 by taking account of any change in SH's required rate of return since the repricing as well as statistical elements.*
- *Points to remember are that*
 - *Risk discount rate should always be greater than the risk free rate, any change in risk free rate, risk discount rate will also change*
 - *Margin between the risk free rate and risk discount rate should reflect all sources of risk*
 - *risk discount rate used in the pricing of different products should reflect their relative risk.*
 - *Overall return earned on the whole capital of the company, generated from its entire business activities, needs to meet the required rate of return (cost of capital)*
 - *Risk discount rate is simply a number that is used as a criterion in profit testing. It is set to ensure that the rate of return is satisfactory.*

Profit Criteria needs to be met from the price charged for a prod.

Consistency

- *Assumptions need to be considered in their totality, not in isolation i.e., making sure we capture the way that variables behave together, where correlation exists between them.*
- *Investment return and inflation : Long term relation of different types of assets return and expense inflation must be valid. However, no need to make assumptions about future investment condition-when there is no risk of reinvestment risk.*
- *Tax: Tax treatment should be reflected in investment return and expense assumption if taxed as (I-E). In profit testing, the profit should be measured net of tax*
- *NB and expense: In case of new product, the amount of loading should be such, given the anticipated volume of sales, total loading will at least recover the development expense, pay a fair share towards the company's overheads, and make additional contributions to profits.*
 - *IR and bonus should be consistent in situations where bonus needs to be loaded.*
 - *Persistency - can be affected by the general economic climate, which will be one of the major factors behind IR assumption. It may also be affected by the bonus level, discontinuance terms and renewal commission levels*
 - *Basis/assumptions should be consistency with related prod.*

10. Explain how can their be circulatory problem

Future volume is a function of competition, which is a function of price. Price is a function of expense loading, which is a function of future volume, therefore we can say, future volume is a function of future volume

11. New WP, addition to benefit method, premium basis - BEL and bonus = 50% of the rates currently being declared by the company on its existing business. Explain the effect

- *Might be cheaper, might increase the volume*
- *Bonus capacity will be lower, therefore bonus will gradually decrease as new policies are issued.*
- *Produce inequality between old and new generations of PH, assuming the company decides to pay an average rate into the two classes of business even though their contribution to profit might be quite different.*

- So either company has to live with this inequality or declare different bonuses for different generations.

12. What might happen if a company doesn't use consistency?

- Might end up selling two fairly similar products, at the same time, of which one must be expensive relative to the other. If market is competitive, this might affect consumer's buying and discontinuance behaviour
- More cheaper product might be sold
- Holders of the most expensive product may lapse their existing policies and take out new cheaper ones, to the disadvantage of the company
- Policyholders could feel aggrieved at the perceived inequality, which resulting bad publicity for the company, and ensuring marketing risk
- The company's profit will become more sensitive to the type of business sold.

Setting Assumption (2)

- Purpose of assumption setting – Published accounts (IFRS), Pricing, reserving, profitability measure, Internal accounting
- Published accounts –consistent with the legislation and accounting principles. Matters to be considered includes:
 - Account preparation should be going concern basis or break-up basis
 - Should show a true and fair value
 - Reserves are required should be assessed as best estimate or on some other basis and precisely how the terms used are to be interpreted
- Supervisory reserving (Insurance regulatory) –
 - Rules governing solvency requirements may or may not be the same as published accounts.
 - May need to be prepared on a going concern or break-up basis.
 - Rules in general are more prudent.
- Internal management account –
 - more realistic assumption.
 - For internal management regarding the decisions to produce expected values of the future experience.Embedded values –

Best estimate reserves

- Purpose – internal/published management account, basis best estimate
- Use cases – business to be sold or rewards to management.
- All experiences (including -ve reserve) will be allowed.
- *Cashflow method would estimate the amounts that need to be held now to meet future claims, expense and tax outgo against the expected premium and investment revenue inflow.*
- *Gross premium formula (Premium = Expected present value of (Benefit+expense+commission+profit) does not take account of withdrawloss. Losses or profit from withdrawal will affect the actual value of the business to the company, and will lead to error unless such profits and losses are expected exactly to cancel out. Therefore this method will be unrealistic and inferior to the cashflow approach, for valuing non-linked products for this purpose.*

1. Discuss whether the effect of underwriting should be allowed in the best estimate reserve?

- In underwriting, life selected will have lower mortality than ultimate mortality for the first few policy years. Therefore, when we calculate reserves, there will be some policies which have recently been enforced and experience less mortality than others.

Valuing life insurance contracts: embedded value

Calculation of embedded value

Embedded value = PV of future shareholder profits + release of shareholder owned net assets.

- The SH owned share of net assets, where net assets = Asset - liabilities (Reserving basis).

These assets may be valued at market value or may be discounted to reflect 'lock in', for example if they are required to be retained within the fund to cover solvency capital requirements.

These net assets may have to be invested cautiously to ensure that the solvency capital requirements are met. As a result, they will be expected to achieve a lower return that could be achieved with a less constrained investment strategy, and so they will be worth less than their market value to the shareholders.

Eg – An insurer holds 100 for solvency capital. Capital is assumed to be locked in for 10 years. 4% is earned on solvency capital and the embedded value is calculated at 5%

The Discounted value of solvency capital = $100 \times (1.04^{10} / 1.05^{10}) = 90.874$

So these assets maybe included in the embedded value, however instead of market value of 100, they are rather included as discounted value of 90.874

- The PV of future SH profits arising on existing business. (Projection Basis)

The process of determining this amount is similar to profit testing, i.e., future cash flows on existing business are projected and used to estimate the future SH profit. These profits are then discounted back at an appropriate rate to determine their PV.

The calculation of this will differ for different types of business.

- Conventional NP – PV of future premiums + investment income - claims - expense + release of supervisory reserves
- UL – PV of future charges – expenses – benefit in excess of unit fund + investment income earned + release of non-unit reserve
- WP – PV of future SH transfers which is calculated either as a proportion of cost of bonus or as a charge deducted from the asset share.

Cost of bonus approach – 90% of surplus is with PH and 10% with SH. Insurers may also decide to hold back some as a terminal bonus, so SH will have to wait until a bonus is declared to receive their transfer. So embedded value is based on the timing of bonus rather than the timing of emergence of surplus.

Deduction from the AS – is same as Unit linked

For NP – SH transfer is release of any margins within the supervisory reserves relative to the assumption within the embedded value calculation, i.e., profitability of reserve

2. Describe, by considering the two components of an embedded value, the mortality risk that a company faces on:

- regular premium NP WL contracts taken out thirty years ago by men aged 30.
 - regular premium WP WL contracts taken out in the last year by men aged 60.
- In a) reserve will be quite large, thus death will cause relatively small reduction in the net assets. The loss of future profits should also be relatively small as there are fewer years left in which margins can be released (profit has already been released from the first 30 years of the contract).

- In case (b), the reserves will be quite small, so each additional death can cause a substantial reduction in net assets. The loss of future profits will also be large – in fact all the expected profit will be lost as the policyholder has not survived the first year.
- 3. WP endowment – 2 year left to maturity. Cash flows = Prem – exp +IR-claims. At the end of 1st year =15 and 2nd year = -100. Current reserve is 100. And by the end of 1st year, insurer expects it to be 110. Insurer expects the return of 4% on its reserves. The insurer has assets of 120. Calculate the embedded value at 8%
 - The embedded value has 2 parts, i.e., net assets and PV of future SH profit
 - Net asset = $120 - 100 = 20$
 - SH profit = interest on reserve + cashflow – increase in reserve = $100 \times 0.04 + 15 - 10 = 9$
 - SH profit at the end of 2nd year = $110 \times 0.04 - 100 + 110 = 14.4$
 - PV of SH profit = $9/1.08 + 14.4/1.08^2 = 20.679$
 - PV of embedded value = net asset + PV of SH profit = 40.679

Appraisal Value

embedded value + goodwill (use its expertise and brand to add value to the business through sales of profitable new business in future too).

Assumptions for calculating embedded value

- Bases in order to calculate an embedded value: a reserving basis and a projection basis.
- We need to know the reserves now in order to calculate the net assets and we need to know the reserves in the future to calculate PV of future SH profits.
- We will also calculate reserves for supervisory reserving basis.
- Projection basis that projects the company's experience into the future. This basis will be used in the calculation of the present value of future shareholder profits, and has no impact on the shareholder-owned share of the net assets. The company may wish to use a different projection basis depending on the purpose for which the EV is being calculated.
- For internal management and published account – best estimate or realistic view
- Appraisal value – realistic assumptions without margins for sale value and for purchase value, more cautious assumptions with margins

Allowing for risk

- EV should have margins to allow for the unpredictability of profit emergence.
- Risk discount rate will reflect that
- Stochastic or market consistent approach

Difference between the Best estimate valuation and embedded value

- Best estimate – present liability on realistic assumption. The sum is compared with total asset value to give realistic solvency.
- Embedded value – consider the cashflow across the portfolio in each time period, rather than PV of cashflows for each policy. The focus is SH profit calculations. It takes account of the cost of capital in the value of SH transfers, because each year's transfer has to be made from profits that arise in excess of supervisory reserve held.

Consistency

Consistency in setting a valuation basis

- Start point -basis used in previous valuation. Deviation from that must be justifiable, i.e., evidence
- Changes in basis have significant impact on the published financial results of the company.
- Justification to regulatory authorities and that it has not been done just to accelerate the distribution of profits to SH or to improve the reported solvency position.
- Justification to fiscal authorities.
- One of profit or loss should be separately identified in the accounts.

Pricing basis

- Allow the cost of setting up supervisory reserves, in the pricing models used.
- it is essential that the assumed reserves in the pricing basis are consistent with the company's actual reserving basis for the contract.

Consistency in setting an embedded value basis

- start point – previous basis
- Change should be justifiable especially when EV is being used to report externally on a company's real worth
- Effect of change must be reported

Pricing basis –

- More Best estimate basis than pricing basis.

Monitoring experience

Reasons of monitoring experience

Experience will be monitored as a part of the control cycle to: (DUMMMP)

- Develop earned asset shares
- Update assumption as to future experience
- Monitor any trends in experience
- Monitor actual compared to expected experience and take corrective actions as needed
 - Make more informed decisions about pricing and about the adequacy of reserves
- Provide management information to aid business decisions.

actions that might be taken if monitoring shows actual experience being worse than expected - RRRC

- i) Re-price products
- ii) Re-design products
- iii) Change the investment strategy, sales strategy, reinsurance strategy, underwriting strategy, profit distribution strategy
- iv) Re-organise the workforce and IT systems to make more efficient use of expensive resources

Data required

Data grouping

- Basic requirement is that there is a reasonable volume of stable, format and flexible data, from which future experience and trends can be deduced.

By consistent we mean data should be in familiar form, i.e., extracted from the same source, grouped according to the same criteria and equal in terms of reliability.

Eg - age for one group is calculated as 'calendar year of death minus calendar year of birth and for other group it is calculated as 'age large birthday at death'.

- The data needs to be divided in homogeneous risk groups, according to the relevant risk factors. However, this idea must be balanced against the danger of creating data cells that have too little data in them to be credible.
- Period over which the data will be collected, i.e., time interval to be analysed.

seasonal influence need to be captured and not be too great because we want to act as quickly as possible on any recent changes - recent experience not an average of recent and ancient so this will place an upper limit on the time interval. Also it will depend on what we are investigating - expense - 1 year and mortality - 3-5 years

- Level of detail in the classification of the data depends upon the volumes of data available. However, it will be desirable to have separate classes of contract. It is because If you have millions of policies, you have enough statistical credibility to split the data finely. Regardless of how small your data set is, mixing different contract classes distorts your analysis.

Big Data

Benefits

- *analysis of large volumes of data to look for patterns and predict future behaviour*
- For example bancassurance (companies which offer financial services that encompass both banking and insurance operations) hold a large quantity of information about a life insurance customer who also has a bank account with them, including data on that customer's spending habits. The bank can use this data for behaviour indicating low mortality rates or high mortality rate and pass this information to insurance operations.
- As more analytical techniques develop it will become more *important to use such information to understand the underlying risks.*
- *allowing more accurate rating for each individual customer.*
- *greater ability to select the preferred risks, i.e., not only be used for underwriting process but also to target the kind of people company wants to insure.*
- *Monitoring the available data to drive better experience*, through early identification of changes in individual insured risks or potentially through being able to intervene and influence customer behaviours. Eg - individual who shows decline in the purchase of fruits and vegetable, insurer could send out advice on healthy eating

Risks

- Reputational damage

- Regulations governing the data might change
- Inaccurate, incomplete or irrelevant data
- Model risk - complex models are used for big data.
- Expense of collecting and analysing the data may outweigh the benefits
-

Analysis of experience

Subdividing the data and factors through which experience can be analysed -

<i>Mortality</i>	<i>Persistency</i>
<i>Type of contract</i>	<i>Type of contract</i>
<i>Age and gender</i>	<i>Age and gender</i>
<i>Duration from entry</i>	<i>Duration from entry</i>
<i>Distribution channel</i>	<i>Distribution channel</i>
<i>Location</i>	<i>Economy</i>
<i>Occupation</i>	<i>Occupation</i>
<i>Medical/Non-Media</i>	<i>Frequency and size of premium</i>
<i>Smoker/NonSmoker</i>	<i>Payment method</i>
	<i>Original term of contract</i>
	<i>Competition</i>
	<i>Perceived value of the product by costumer</i>

If regulation does not permit the use of sex (or gender) as a rating factor, we would still analyse male and female rates separately if possible in order to calculate the unisex rate that reflects our balance of male and female lives.

Persistency experience

In practice, only type of contract, distribution channel and Duration in force will be considered for grouping to protect ourselves from over homogeneous grouping

Persistency rates and distribution channels -

- Lowest - direct sales force.
- Medium - direct marketing because PH would have initiated the sales and might lapse later because they realise they don't need the policy.
- Highest - broker.

Process

For each homogeneous group, full withdrawal rates can be determined as follows:

- *First year persistency rate can be calculated,*
no of contracts that inforce contract at the end of year/number contracts issued in previous years, ignoring maturity and mortality
- *Withdrawal rate = 1 - persistency rate*
- *Similar procedure can be used to obtain persistency rate and withdrawal rate for 2nd, 3rd etc*
- *The other way to calculate persistency rates cumulatively as the number surviving the year divided by the number in force at the outset of the contract.*
- *Analyse the results and see if experience is inline with expectation or not. If not reason why with accuracy*
- *After finishing the analysis to the greatest possible degree of accuracy, the presentation the information to management according to the requirement and how they want the results. Eg - management would want to see the results split by broad product class not by every product*
- *Perform the analysis on policies being made paid-up or partial withdrawal rates*

Expense experience

- Expenses are the function of the work that the company does
- Analysis will depend on the Purpose and desired end results
- For total profit - sufficient simply to know the expenses in total
- Contribution method - analyse group expenses so that the dividend can be properly calculated for each group.
- Asset shares , terminal bonuses/dividends, or SV scales - appropriate historical expenses between different policy types
- For pricing and reserving - policy's fair cost to the company, so that the correct premium or charges can be levied
- company's future expense experience- determine relationship between the company's business - the policies it sells and maintains in force and the expenses incurred in doing so
- *Expense need to be subdivided as follows:*
- *Expense relating to new business, existing business, termination, investment management*
- *Direct (expense attributed directly to a policy) and overheads (fixed expense independent of new business or policy administration)*
 - *Direct expense are typically variable but maybe fixed to some extent like underwriting cost, but the salary of underwriters is fixed irrespective of business because we won't hire or fire individuals on daily basis, so it is fixed for atleast short term*
 - *Similarly, overheads are fixed cost but will be variable to some extent. Eg - the cost of head office building is independent of how many policies are sold in a month but companies with large business will have much larger head office then small companies*
- *Identify weather expense are:*
 - *Proportionate to the premium payable under the policy*
 - *Proportionate to the SA*
 - *Fixed amounts per policy*
- *Time period used for an expense investigation will depend on purpose*
- *Generally annually*

- Any shorter period will not capture the seasonal influence of one expense eg - NB
- Companies would look at 3 to 5 years of expense experience to identify and if necessary, allow for unusual fluctuations and identify trends over time.
- adjust the figure for inflation, so that cost can be compared in real terms.
- Commissions are generally excluded from expense analysis because it is generally known and thus can be loaded in the premium.
- So we are only concerned about non-commission expense which can be split as

Breakdown of non-commission expenses	When the expenses arises	Any other factor
Initial	Start of the policy	Is the expense proportionate to the: - No contract written or in force? - Amount of benefit written or inforce? - Amount of premium written or inforce?
renewal	Regular during the policy term	
termination	End of the policy	
investment	Management of the company's asset	

Generally expenses are proportional to the number of contracts written or inforce. Exceptions are:

- Marketing expense - may be related to the amount of initial commission paid.
- Underwriting expense - may be related to the size of benefit
- Investment expenses are normally expressed as a percentage of funds under management so that they can be treated as a deduction from an earned investment return.

Types of expenses -

Salary and salary related expenses

- Staff can be split into three groups:
 - Single cell staff - 100% dedicated to one product/activity, i.e., someone who does one specific job for 1 specific product (Direct expense - easy to allocate, dump woe salary in 1 cell)
 - Multi-cell Staff - Works across several specific products - Customer service reps, underwriters, or claims handlers (Direct expense - time sheets)
 - Other staff - Corporate support / general overhead - People who keep the company running as a whole (e.g., HR, Payroll, Legal, Building Security, CEO). (mix - contains both **direct operational tasks** that support policy processing and **general corporate overheads**.)
- Because there is no perfect mathematical formula to separate them, actuaries split and allocate these costs using a two-step pragmatic approach.

Step 1- separate direct and overhead expenses

Direct Operational Component: Work that directly facilitates selling, underwriting, or servicing policies and split them further in group i and ii

Pure Overhead Component: Corporate maintenance that cannot be linked to any policy or product line.

- Example: An IT technician configuring the Chief Executive's laptop.
- Example: Catering and cafeteria staff preparing daily office meals.

Step 2- Apportion the Remaining Overhead Component - Approaches to allocate this cost -

- splitting them down in proportion to the direct expenses already identified - eg - If Product A consumes 70% of all direct staff salaries and expenses across the company, Product A is automatically assigned 70% of the head-office rent, cafeteria costs, and executive salaries.
- splitting them in proportion to expense charges from the policies - If Unit-Linked policies generate \$5M in Annual Management Charges (AMCs) and Term Assurance generates \$1M in premium expense loadings, overheads are apportioned in a 5:1 ratio. This ensures that the products generating the most expense revenue absorb the bulk of company overheads.
- splitting them down in proportion to something else - eg - new business or renewal etc staff they serve - eg - **Catering / HR Costs:** Apportioned based on **departmental headcount** (if the New Business department has 80 employees and the Renewal department has 20, catering costs are split 80:20).

Property costs

- Owns buildings - notional rent needs to be charged to the relevant departments because it is costing the rent to the company it would have been earning otherwise.
- This rent, plus property taxes, heating, lighting, cleaning costs etc, can be split, for example by floor space occupied, between departments and then allocated in accordance to salaries.

Computer costs - amortised over its useful lifetime and then added to the ongoing computer costs. These can then be allocated according to computer usage.

Investment costs - reduced in yield

Process

- We start by knowing the expense such as salary cost, computer cost etc
- Commission will be excluded from the analysis
- Subdivide the non-commission cost into the required 'cells' or homogeneous groups - into initial, renewal, terminal and investment and then 1st 3 will be further grouped into whether they are proportional to the number of contract, amount of benefit or premium
- Some expenses can be allocated directly to a particular cell
- Staff expenses may need to be subdivided between cells by the use of timesheets
- Overhead expense needs to be allocated pragmatically eg - proportion to the split in direct expenses
- Property expenses can be allowed for by charging a notional rent to each department in proportion to the floorspace occupied, then allocated to the different cells in proportion to the department's salary costs.
- Cost of new computer equipment can be spread over their future expected lifetimes and then allocated to departments in proportion to usage.

- *Investment costs would be subdivided by asset class and usually allowed for by a reduction in yield for each class.*
- *One-off capital costs would be spread over the expected future lifetime of the item, then just treated as an overhead.*
- *Exceptional expense items may be ignored in the analysis, but their future incidence may be allowed for in margins in the future expense assumptions or risk discount rate.*

1. Outlines checks that could be made at this stage.

- Compare results with those of the previous expense investigations to look for trends which may indicate the need for action.
- Compare result with the industry stats for checks and possible problem area.

Analysis of surplus/profit

A company will want to analyse its surplus in order to: (DIVERGENCE)

- **Divergence of A vs E**
- **Information to management and for accounts of trends in the experience of the company**
- **Variance of whole = sum of variance of parts (i.e., explain the total change in surplus over the years including immaterial unexplained)**
- **Experience monitoring to feedback into acc**
- **Reconcile values for successive years**
- **Group into one-off/recurring sources of surplus**
- **Executive remuneration scheme (data for), and assistance with bonus distribution to identify non-recurring (one time event due to which surplus has raised or loss occurred) components of surplus, thus enabling appropriate decisions to be made about the distribution of surplus to with-profits policyholders. This is mainly helpful when bonus is at discretion of the company because for the revalorisation method - a pre-defined method is used, therefore analysis won't be required.**
- **New business strain *i.e., show financial impact of new business**
- **Comply with regulatory requirement**
- **Extra check on valuation data and process, if carried independently - If we derive the analysis of surplus components from scratch (rather than working backwards from the valuation result) and they total to give the surplus arising then we have performed a valuable check on the valuation process. Even if the analysis is not done with completely independent data, it can still highlight errors, and so has a useful checking function**

Valuation surplus arising

- Change in the company's surplus over the year
- $\text{Surplus} = \text{asset} - \text{liabilities}$ then $\text{Change in surplus} = \text{change in asset} - \text{change in liabilities}$

Financial effect of Divergence of A vs E

- This will either result in surplus or loss
- tell us its financial impact
- New business will impact negatively on surplus because cost in the initial year is more than the company has received as premium especially in case of regular premium

Analysis of embedded value profit

- Change in embedded value = change in (SH owned net asset + value in force(PV of expected future profits from existing business))
- Uses -
 - *Validate the calculations, assumptions and data used*
 - *Since it is same assumption used for pricing and embedded value calculation (except for RDR which is often set higher for pricing to reflect the risk), it is crucial they are correct and their continuing validity can be monitored by embedded value*
 - *Reconcile the values for successive years*
 - *To protect ourselves from the modelling error.*

Eg - consider a policy portfolio is the giving a profit of 10pa for the next five years, free assets = 100 and RDR=10% and asset earn 8% then

Present value of Future profits $\ast PVFP = 10a_5 \bar{\mid} = 37.9$

So at the start of the year we have, embedded value = free assets + PVFP = 137.9

So at the end of the year, free asset will change to 108, as they will earn the interest, PVFB will left for 4 years so it will become $10a_4 \bar{\mid} = 31.4$, therefore Embedded value = 149.7

Eg - Impact of new business on Embedded value

PVFP = 5 before any cost incurring from NB, Free asset 100, initial cost and commission=10 and premium = 3

Embedded value after sale before cashflow = 105

After incurring cost, strain of $10-3=7$ will be given from free asset and then it will become part of the PV of future profit

=> PVFP=12 and free asset = 93=> embedded value = 105

- *Provide management information. The sort of information it will give is:*
 - *Value of new business, normally by prod*
 - *Amount of any expense, mortality and withdrawal profit or loss*
 - *The impact of free asset on embedded growth*
 - *The impact of SCR on return achieved*

Why we need to separate embedded value analysis for expense, mortality and withdrawal experience when already done in surplus analysis

- *Embedded value is calculated on BEL - look at the PV of future profit unlike in surplus only free assets are used*
- *Provide data for executive remuneration*
- *Provide detailed information for publication in the company's accounts or those of any parent company, in particular the value of new business taken on by the company*

Using the results (PRRICE)

- ***Providing the adequacy of staffing***
- ***Revising product design, underwriting process, reinsurance arrangement, claim handling processes, sales procedures in term of training, selection of distribution, wording and format of sales literature and the mechanics of any commission payments and clawbacks (PURCSDLC)***
- ***Raising capital and/or catering the capital allocation methodology if surplus is negative***
- ***Improving the pricing basis, actuarial model, risk management and governance process, contract wordings, system and data recording processes, retention activity to improve persistency experience***
- ***Changing the product mix / launching new business, investment strategy, surplus distribution approach, marketing message, target market and/or distribution channel***
- ***Establishing and revising the reserving basis***